

Market System Development - Status and Pre-Research Review

Version 3 - Research R.9E

Generated from: custom_terminal main commit 450e976ae472fa440a704c74ad959b60f1113219
Research-state fingerprint: f7978db65c011c0dccf43dfd94623528a2e9672cc40c8c8505cb7666d6a69f38
External reference: version2.0 master commit f9a6eaec2cab1dd9e85d284e48b9863cae0b1298, tree
ad3c21fb2244f0acd7680bd0bdc4958d2516b16f Review state: REPORT_GENERATED_PENDING_REVIEW

Covered next scope: EXACT_PRODUCTION_LOCATOR_BINDING_PREPARATION

NO SQLITE CONNECTION, SQL, PRAGMA, AUDIT EXECUTION, MARKET-ROW ACCESS, MARKET ANALYSIS, SIMULATION, BACKTESTING, SCORING, RECOMMENDATION, BROKER ACTION, OR TRADING IS AUTHORIZED BY THIS REPORT.

Executive summary

This project is building a personal Indian-market research and capital-allocation support system whose conclusions can be reconstructed from point-in-time evidence. The intended system preserves data provenance, calculates deterministic features and outcomes, tests hypotheses out of sample, measures uncertainty and costs, and may eventually support portfolio and downside-risk decisions. It is analysis-only. It has no approved investment edge, production score, recommendation, or trading capability.

Research governance comes before strategy testing because a plausible backtest can be invalidated by survivor selection, unstable identity, revised data, incorrect event timing, ambiguous execution, or arbitrary score weights. The system therefore separates raw evidence, deterministic calculations, validation, confidence, decisions, and portfolio risk. AI may interpret evidence and generate hypotheses, but it is not the quantitative source of truth.

Infrastructure is the strongest completed result. Through R.9D, the repository has provider-neutral data contracts, immutable evidence patterns, a governed research lifecycle, an independently audited synthetic canary, a synthetic-only bounded F&O auditor, and a durable production-enablement boundary. The boundary includes atomic one-use approval consumption, append-only hash-chained events, crash-detectable attempts, bounded evidence retention, secret and path redaction, and network and broker isolation.

Production access is still impossible. The production locator remains disabled, the actual configuration value has not been read, the real database path has not been resolved, the database has not been located or opened, and its identity is unknown. No production registry, real approval, or audit attempt exists. Synthetic safety tests do not establish the trustworthiness of the real database.

This report asks the owner to approve only EXACT_PRODUCTION_LOCATOR_BINDING_PREPARATION. If explicitly approved later, that phase may perform one bounded filesystem-only identity ceremony and prepare a sanitized binding proposal. It may not connect to SQLite, execute SQL or pragmas, inspect schema or market rows, create or consume an audit approval, or execute the audit.

Current status at a glance

Area	Status	Meaning
Architecture	ESTABLISHED	Trust, research, evidence, and application responsibilities are separated.
Historical equity data	NOT TRUSTED	Population, identity, inactive listing, and terminal-outcome evidence remains incomplete.
Momentum 12-1	NON-ACTIONABLE	Reproducible infrastructure exists; evidence gates remain closed.
Research governance	SYNTHETICALLY PROVEN	A synthetic canary completed and was independently audited.
F&O auditor	SYNTHETIC ONLY	Bounded Stages 1-3 work on temporary synthetic fixtures only.
Durable production boundary	IMPLEMENTED BUT DISABLED	Governance exists; exact production binding and approvals do not.
PDF v3	PENDING OWNER REVIEW	This report cannot authorize its own covered next phase.

1. What are we building?

The objective is a trustworthy research machine first and a decision-support product later. Every important score should represent one or more declared relationships between information knowable at decision time and a later outcome. A credible score should disclose historical bucket behavior, out-of-sample and walk-forward performance, sample size, transaction-cost sensitivity, drawdown, predictive strength, regime dependence, decay, confidence, and provenance.

The target evidence chain is:

```
Data sources
-> immutable raw and point-in-time normalized evidence
-> deterministic features and declared outcomes
-> preregistered experiments
-> walk-forward validation
-> evidence and confidence
-> decision policy
-> portfolio and risk support
```

Prediction quality, economic quality, and portfolio quality remain separate. A statistically interesting relationship is not automatically executable. A profitable historical result is not trustworthy if the eligible population was reconstructed from survivors, identifiers were guessed, delistings were omitted, corporate actions were destructively adjusted, or costs used the wrong effective date.

The AI layer is an interpreter and research assistant. It may explain results, propose hypotheses, and suggest falsification tests. Deterministic and statistical code must own data transformations, features, outcomes, backtests, scores, confidence, and portfolio risk.

The system remains analysis-only. Broker actions and trading are prohibited.

2. Why governance precedes strategy testing

Research iteration creates incentives to change a hypothesis after seeing its result. A governed system counters that risk with explicit definitions, immutable inputs, locked specifications, out-of-sample evaluation, one-use approvals, manifests, and separate lifecycle decisions. This does not guarantee a valid edge, but it makes hidden changes and unsupported promotion easier to detect.

Infrastructure evidence and market evidence are different. Passing concurrency, hashing, or manifest tests shows that the machinery behaves as specified on tested fixtures. It does not prove that historical prices, security populations, corporate actions, costs, benchmarks, or F&O rows are complete and point-in-time correct.

3. Milestones through R.9C

Architecture and Vertical Slice A

- The original Streamlit scaffold was audited and reorganized conceptually around shared engines instead of duplicated page calculations.
- Vertical Slice A migrated the 12-1 momentum experiment into typed temporal data, feature, outcome, universe, cost, experiment, manifest, and walk-forward contracts.
- A deterministic golden fixture preserved compatibility evidence. It is a debugging anchor, not statistical proof.
- The experiment remained non-actionable and was not converted into a production score.

Historical-data qualification

- Slice A.5 found that the local equity panel was not trustworthy for cross-sectional research because the security population and inactive listings were incomplete.
- A.6 introduced provider-neutral ingestion, immutable raw manifests, typed normalization, identity reconciliation, corporate-action handling, and dataset acceptance.
- A.7 and A.8 qualified public official evidence conservatively. Bhavcopy availability was not treated as a historical security population.
- Only 3 of 12 locked historical security/bhavcopy pairs qualified. Historical population reconstruction remains incomplete.
- A.9 and A.10 preserved official-response ambiguity instead of inferring permission.
- A.11 and A.12 established bounded read-only current Kite coverage. Current instruments are not a historical universe.

Governed execution and independent evidence

- R.1 and R.2 preserved legacy evidence as noncanonical rather than rewriting it.
- R.3 and R.4 introduced research families, preregistration, declared inputs, split access, exact approvals, execution events, manifests, evidence import, and lifecycle controls.
- R.5 executed one tiny synthetic infrastructure canary under an exact one-use approval.
- R.6 independently audited that canary and confirmed the governance chain, not a market edge.
- R.7 established the pre-research PDF review gate and reconciled the external version2.0 repository as separate and reference-only.

Planning and the synthetic F&O auditor

- R.8 kept market sentiment and stock-score concepts separate because their economic meanings, clocks, populations, and missingness differ. No combined score or arbitrary weights were created.
- R.9A defined a tightly bounded, hashed F&O audit proposal limited to file identity, SQLite catalog safety, and local provenance inventory.
- R.9B implemented the auditor with synthetic fixtures only. It prohibited market-table reads and rejected the production locator.
- R.9C generated PDF v2. The owner later approved only production-enablement design and synthetic testing. That approval is preserved historically and became stale when R.9D changed the governed source state.

4. Repository authority and current assets

`custom_terminal` is the canonical authority for provenance, point-in-time trust, research definitions, approvals, execution evidence, lifecycle, and promotion. `version2.0` remains a separate product and reference repository. Its dashboards and ideas may inform later design, but its calculations are not canonical evidence.

The public `version2.0` master branch was rechecked read-only for R.9E on 2026-09-03. It remains at commit `f9a6eaec2cab1dd9e85d284e48b9863cae0b1298`; therefore its bound tree remains `ad3c21fb2244f0acd7680bd0bdc4958d2516b16f`. It has not changed from the previously reviewed binding and was not cloned, copied, or executed.

Capability	What exists	Evidence boundary
Temporal and provenance contracts	Observation, publication, effective, retrieval, revision, source, parser, and quality concepts	Can express point-in-time state when source evidence exists.
Immutable evidence patterns	Raw hashes, manifests, content-addressed objects, and artifact inventories	Detects mismatch against preserved anchors.
Research governance	Families, preregistration, split controls, exact approvals, and lifecycle	Prevents informal promotion into canonical evidence.
Synthetic canary	One completed, independently audited run	Proves a narrow infrastructure path on non-market data.
Kite current data	Memory-only login and bounded read-only snapshots	Current display only; not historical research evidence.
Synthetic F&O auditor	Bounded identity, catalog, and provenance inspection	Tested only on temporary synthetic SQLite fixtures.
R.9D production boundary	Durable approval and evidence governance with fail-closed locator	Production remains deliberately unreachable.

No displayed score or recommendation has passed the required evidence lifecycle. Sentiment and stock scores remain future candidates, not approved calculations.

5. What remains untrusted?

Capability	Status	Consequence
Historical listed population	FAIL / INCOMPLETE	Traded rows cannot stand in for all eligible securities.
Survivorship safety	FAIL	Missing inactive or failed listings can bias results.
Stable identity history	INCOMPLETE	Symbol, series, ISIN, merger, and listing transitions remain unresolved.
Economic terminal outcomes	FAIL	A final quote is not authoritative delisting consideration.
Corporate actions	UNKNOWN / PARTIAL	Price discontinuities do not establish action type or total return.
Benchmark PRI/TRI	PARTIAL / UNKNOWN	Price-return evidence cannot be silently labeled total return.
Historical statutory costs	INCOMPLETE	Unknown intervals must fail rather than inherit current rates.
F&O database identity	UNKNOWN	No production database byte or path has been inspected.
F&O provenance and retention	UNKNOWN	File size and apparent span do not prove source lineage or rights.
F&O market-row quality	NOT EVALUATED	Coverage, values, and point-in-time fitness remain outside Stages 1-3.

6. The R.9A and R.9B audit boundary

R.9A permits only three future audit stages, subject to later exact authorization.

Stage	Permitted purpose	Explicit exclusions
1	Bounded target identity	No full hash, copy, export, or market interpretation.
2	SQLite read-only safety and catalog metadata	No user-table rows, aggregates, coverage, or integrity scan.
3	Already-local provenance and retention inventory	No network, acquisition, parser execution, or market-data reads.

Stages 4-6 remain excluded. `quick_check`, `integrity_check`, market-table row reads, dates, symbols, expiries, strikes, prices, volume, open interest, returns, features, scores, strategies, recommendations, and trading remain prohibited.

Exact R.9A bindings

Object	SHA-256
Proposal package content	9eff345b453d1f0f0072c927f2a7dcb5b60cd98af6d9c415703fa0e504016acd
Proposal manifest	4fe0091ac405fbd5eac027038f6350b3b6a8bbe88d9141a982516b6489e95ed8
Audit scope	86af643f9c88a1162ecaecf76d22e4e74b5d39816f6e6117eb6105b726588d8d
Resource envelope	1aec4f54a6e3cf88cfa1ecde91b5b8c1579886d38a5282e5131471b2694436b5
Expected outputs	b1d05cfba2de056156d07c4f93512372af9eafd6f2de7d682404050e742f0d8a

R.9B uses SQLite URI `mode=ro`, verifies `query_only`, disables extension loading, enforces a statement allowlist and authorizer denylist, limits statements, applies a monotonic progress deadline, redacts statements, detects sidecars and mutation, bounds provenance traversal, restricts outputs, and finalizes artifacts atomically. It consumes its exact approval before the first target connection.

The auditor has only been executed against small temporary synthetic fixtures. No result from R.9B qualifies the real database.

7. R.9D durable production boundary

R.9D replaced the process-local production-enablement concept with durable synthetic-tested governance while keeping the real target unreachable.

Durable approval and event model

- A separate local-data-audit approval type prevents substitution of a market-research approval.
- Approval registration preserves the exact canonical payload hash and rejects duplicate IDs or payloads.
- Events are append-only and linked by SHA-256, making reordering or mutation detectable against the event head.
- Approval consumption uses a SQLite BEGIN IMMEDIATE transaction and unique constraints.
- Two spawned processes raced for one approval; exactly one succeeded and the other failed before target connection.
- A consumed approval remains consumed after process restart.
- A crash before committed consumption leaves the approval unused.
- A crash after committed consumption leaves a durable incomplete attempt for later inspection.
- Completed, aborted, and failed terminal events reconcile against the attempt projection.
- Approval consumption occurs before any target connection.

Sanitization and retention

- Private absolute paths and secret-like values are rejected or redacted from durable governance events.
- Runtime dependency injection is rejected at the production boundary.
- The module has no network, broker, Streamlit, scoring, portfolio, recommendation, or trading call path.
- Synthetic audit outputs are written under a task-scoped store separate from source data.
- Output names and total bytes are bounded, hashes are exact, publication is atomic, and finalized attempt directories are immutable by contract.
- Retention metadata states that restore was not tested and no backup is claimed.
- Synthetic artifacts remain noncanonical and ineligible for promotion.

These controls are application-level and tamper-evident. They are not physical protection against the machine owner, and they do not establish real database quality.

8. The deliberate production interlock

Production access is still impossible.

The locator state is:

```
PRODUCTION_LOCATOR_DISABLED_PENDING_EXACT_BINDING
```

The deliberately impossible R.9D interlock is:

```
R9D_EXACT_PRODUCTION_ACTIVATION_IMPOSSIBLE
```

Current facts:

- The configuration value has not been read.
- The real path has not been resolved.
- The database has not been located.
- The database has not been opened.
- The database identity is unknown.
- No production registry exists.
- No real audit approval exists.
- No real audit attempt exists.
- The activation template is unsealed, unregistered, unusable, and contains missing binding fields.
- The production entry point fails before configuration reading, path resolution, filesystem identity reads, attempt creation, or SQLite access.

Synthetic tests demonstrate behavior for constructed inputs under the tested runtime. They do not show that the real file is SQLite, that it is stable, that its schema is safe, that its provenance is adequate, that its rows are complete, or that its market data is suitable for research.

PDF v2 remains byte-exact historical evidence of the owner's earlier limited approval. It is now PDF_V2_STALE_AFTER_R9D_IMPLEMENTATION because R.9D changed the research-state fingerprint.

9. Exact proposed next action

The only proposed next phase is:

```
EXACT_PRODUCTION_LOCATOR_BINDING_PREPARATION
```

Only after explicit owner review and approval of this exact PDF may that phase perform the following bounded filesystem operations:

1. Read the approved configuration file.
2. Resolve only paths .fno_db.
3. Verify that the resolved target is a regular file.
4. Reject symlinks and path escapes.
5. Assign a sanitized alias; keep the absolute path private.
6. Record file size and modification time.
7. Read the 100-byte SQLite header without opening SQLite.
8. Calculate one bounded, deterministic 64-position sampled identity pass.
9. Record the position/chunk hashes and aggregate sampled-identity root.
10. Verify that size, modification time, header, and relevant sidecar state did not change during the pass.
11. Prepare an exact sanitized binding proposal for later owner review.

That phase must not:

- Call `sqlite3.connect`.
- Execute a pragma or SQL statement.
- Create the production audit registry.
- Create, register, or consume an audit approval.
- Inspect schemas, tables, indexes, or foreign keys.
- Read any market row or calculate market statistics.
- Execute audit Stages 1-3.

Approval of locator-binding preparation is not approval of the later audit.

10. Identity cost and limitations

The proposed identity pass is bounded and deliberately weaker than a full-file hash.

Item	Bound or meaning
Deterministic positions	64
Chunk size	4 MiB
SQLite header	100 bytes
Maximum expected read for one pass	268,435,556 bytes
Expected elapsed estimate	Proposal estimate only; depends on local storage and file size
Output	Chunk hashes plus one aggregate sampled-identity root

Sampled identity is not a full SHA-256. Modification time is metadata, not identity. Unchanged sampled chunks do not prove that unsampled bytes are unchanged. The pass does not prove SQLite structural integrity, schema safety, market-row quality, provenance, completeness, or point-in-time fitness.

The absolute path remains private. Committed evidence may contain only a sanitized database alias and non-identifying bounded metadata. No full database copy or export is proposed.

The locator-binding phase should perform one identity pass. It must not quietly consume the five-checkpoint identity budget reserved for a later approved audit. A second bounded comparison may occur only if mutation detection is expressly covered by the same later approved scope.

11. Standing constraints

- Paid data is prohibited; free official sources are preferred.
- Kite Connect may supplement data only where governed and does not solve historical population gaps.
- `version2.0` remains separate and reference-only.
- Sentiment and stock scores remain future candidates.
- Recommendations and live scans remain hidden.
- No trade execution is permitted; the project remains analysis-only.
- Broker actions and trading remain prohibited.
- Display-only provider decisions and legacy-label cleanup remain deferred.

12. Future authorization ladder

PDF v3 generation

- > owner reviews PDF v3
- > owner approves locator-binding preparation only
- > bounded filesystem identity ceremony
- > exact sanitized binding report
- > owner reviews exact database identity
- > production interlock removal proposal
- > implementation review
- > exact one-use audit approval
- > approval consumption
- > first SQLite connection
- > Stage 1-3 audit
- > independent audit review
- > decision whether later stages may be proposed

No step automatically authorizes the next. In particular:

- This PDF is not self-authorizing.
- Locator-binding approval cannot authorize SQLite access.
- A binding report cannot remove the production interlock.
- Interlock-removal implementation cannot create audit authority.
- A one-use approval authorizes only its exact target, stages, envelope, outputs, issue/expiry interval, and attempt.
- Completion of Stages 1-3 cannot authorize market-row reads or Stages 4-6.
- Audit evidence cannot become a market score or recommendation without a separate research lifecycle.

Approval separations

Decision	May permit	Cannot permit
PDF v3 owner review	Confirms summary and exact covered preparation scope	Any operation by itself
Locator-binding approval	Bounded filesystem-only identity preparation	SQLite connection or audit execution
Exact binding review	Confirms the sanitized target identity	Automatic interlock removal
Interlock implementation review	Confirms reviewed code and bindings	Audit execution by itself
Exact one-use audit approval	One bound Stage 1-3 attempt	Market rows, later stages, or research
Independent audit review	Interprets bounded audit evidence	Automatic database trust or strategy approval

13. Current verification evidence

R.9D verification results are infrastructure evidence only:

Verification	Result
Focused R.9D tests	19 passed
Focused R.9C/R.9D boundary tests	30 passed
Combined R.7-R.9D governance	111 passed, 2 skipped
Complete root suite	324 passed, 2 skipped
Complete separate Data-test suite	289 passed
Concurrent consumption	Exactly one of two contenders consumed approval
Production locator non-resolution	Passed
Network and broker isolation	Passed
Protected evidence	Unchanged
JSON validation	84 valid, 0 invalid
Secret-pattern scan	0 credential-pattern hits

The two skipped tests exercise Windows symlink behavior in an environment without unprivileged symlink creation. Direct path-escape and production symlink guards passed, but the platform-specific execution cases remain a documented limitation.

The research-state fingerprint changed during R.9D from 9382d3d7511dba094a1321294d4f6575cf2cd857a0e43c6d0c024df55202ec31 over 231 files to f7978db65c011c0dccf43dfd94623528a2e9672cc40c8c8505cb7666d6a69f38 over 237 files. PDF v3 binds the latter state and source commit 450e976ae472fa440a704c74ad959b60f1113219.

Passing these tests does not prove the real database's identity, provenance, completeness, SQLite integrity, schema, row quality, or suitability for any market hypothesis.

14. What remains prohibited?

- Reading the F&O configuration value before later explicit owner approval.
- Resolving or disclosing the real absolute database path.
- Locating, opening, connecting to, copying, exporting, repairing, or querying the real database during R.9E.
- Calling SQLite, executing SQL or pragmas, or inspecting schemas and tables during locator-binding preparation.
- Creating a production registry, real activation, real audit approval, or audit attempt.
- Consuming any approval or executing audit Stages 1-3 under this PDF alone.
- Reading F&O market rows or calculating dates, symbols, expiries, strikes, prices, volume, open interest, returns, coverage, or integrity metrics.
- Running market analysis, simulation, backtesting, parameter search, ranking, scoring, recommendations, portfolio actions, broker actions, or trading.
- Treating current Kite instruments as a historical universe.
- Treating `version2.0` output, momentum, sentiment, stock scores, or audit output as actionable evidence.
- Acquiring paid data or silently broadening external-source rights.

This report is a review artifact. It does not authorize execution, and none of its pending questions has an answer at generation time.

15. Owner-review questions

Please answer every question explicitly after reading this exact PDF.

1. Is PDF v3 an accurate summary of the project through R.9D?
2. Do you confirm that `version2.0` remains separate and reference-only?
3. Do you authorize only `EXACT_PRODUCTION_LOCATOR_BINDING_PREPARATION`?
4. Do you authorize reading the configured locator value and performing the bounded filesystem identity pass described in the report?
5. Do you confirm that SQLite connection, SQL, pragmas, schema inspection, and market-row access remain prohibited during that phase?
6. Do you confirm that the resolved absolute path must remain private and only a sanitized alias may appear in committed evidence?
7. Do you accept that the sampled identity is not a full-file hash?
8. Do you confirm that a later exact approval is still required before the first database connection?
9. Do you request any correction before locator-binding preparation?

Reviewer answers: **UNRESOLVED** Reviewer identity: **UNRESOLVED** Review timestamp: **UNRESOLVED** Review decision: `REPORT_GENERATED_PENDING_REVIEW`

An acceptable review must identify this report as Version 3, occur after its generation, and answer all nine questions. A prior approval, silence, or a generic instruction to continue does not satisfy the gate.

Appendix A - Milestone timeline and decisive states

Milestone	Result	Evidence class
Architecture and Slice A	Reproducible momentum infrastructure; non-actionable	Research process
Slice A.5	Local equity dataset not trusted	Data quality
A.6-A.8	Provider-neutral ingestion; public population sample incomplete	Infrastructure and source qualification
A.9-A.10	Official-response gate remained closed	Access governance
A.11-A.12	Read-only current Kite coverage ready	Current-market infrastructure
R.1-R.4	Legacy preservation and governed execution controls	Governance infrastructure
R.5-R.6	Synthetic canary and independent audit	Infrastructure evidence
R.7	PDF v1 reviewed for planning only	Owner review
R.8	Sentiment/score ambiguity and F&O audit planning	Definition and planning
R.9A	Exact bounded Stage 1-3 proposal	Data-audit proposal
R.9B	Synthetic-only auditor	Safety infrastructure
R.9C	PDF v2 reviewed for R.9D design/testing only	Owner review
R.9D	Durable production boundary implemented but disabled	Safety infrastructure
R.9E	PDF v3 generated pending review	Documentation/review preparation

Current decisive states

- Historical equity data: NOT_TRUSTED_FOR_CROSS_SECTIONAL_RESEARCH.
- Historical population: HISTORICAL_POPULATION_RECONSTRUCTION_INCOMPLETE.
- Official NSE response: AWAITING_SUBSTANTIVE_OFFICIAL_RESPONSE.
- Kite: CURRENT_MARKET_DATA_ONLY.
- Momentum: non-actionable and not promoted.
- Sentiment definition: SENTIMENT_DEFINITION_AMBIGUOUS.
- Stock-score definition: STOCK_SCORE_DEFINITION_AMBIGUOUS.
- F&O auditor: synthetic only.
- Production locator: PRODUCTION_LOCATOR_DISABLED_PENDING_EXACT_BINDING.
- PDF v2: stale historical reviewed evidence.
- PDF v3: REPORT_GENERATED_PENDING_REVIEW.

Appendix B - Evidence and hash index

Evidence	Binding
custom_terminal summarized source commit	450e976ae472fa440a704c74ad959b60f1113219
Research-state fingerprint	f7978db65c011c0dccf43dfd94623528a2e9672cc40c8c8505cb7666d6a69f38
PDF v2	765c2facad827a3a6473b605037d9975135885e767420ca810ebbc28852c2adf
version2.0 commit	f9a6eaec2cab1dd9e85d284e48b9863cae0b1298
version2.0 tree	ad3c21fb2244f0acd7680bd0bdc4958d2516b16f
R.9A proposal package	9eff345b453d1f0f0072c927f2a7dcb5b60cd98af6d9c415703fa0e504016acd
R.9A proposal manifest	4fe0091ac405fbd5eac027038f6350b3b6a8bbe88d9141a982516b6489e95ed8
R.9A audit scope	86af643f9c88a1162ecaecf76d22e4e74b5d39816f6e6117eb6105b726588d8d
R.9A resource envelope	1aec4f54a6e3cf88cfa1ecde91b5b8c1579886d38a5282e5131471b2694436b5
R.9A expected outputs	b1d05cfba2de056156d07c4f93512372af9eafd6f2de7d682404050e742f0d8a
R.9D locator state	PRODUCTION_LOCATOR_DISABLED_PENDING_EXACT_BINDING
R.9D deliberate interlock	R9D_EXACT_PRODUCTION_ACTIVATION_IMPOSSIBLE

Plain-language glossary

Term	Meaning
Point-in-time	Uses only information knowable at the declared historical decision time.
Survivorship bias	Results are distorted because later failures or disappearances are omitted.
Stable identity	A durable security/listing identifier that survives symbol and name changes.
Preregistration	A locked question, data, method, metric, and decision rule written before outcome access.
One-use approval	Exact authorization consumed by one attempt and never reusable.
Catalog-only audit	Reads SQLite metadata without reading user-table market rows.
Sampled identity	Hashes deterministic byte ranges; bounded change detection, not a full hash.
Canonical evidence	Evidence admitted through the governed approval, manifest, event, and import chain.
Report gate	Owner-review prerequisite; never execution authority by itself.

Current status: REPORT_GENERATED_PENDING_REVIEW.

PRE_RESEARCH_PDF_V3_READY_FOR_OWNER_REVIEW